Splitting Fields, Separability, Algebraic Closure and Galois Theory Basics

From the class book: Chapter 21, Sections 21.4 to 21.7; Chapter 22, Sections 22.3 to 22.5

1. Key definitions

Splitting field. For $f \in F[x]$ of degree $n \geq 1$, an extension E of F is a splitting field of f if f factors into linear factors over E and E is generated over F by the roots of f.

Separable.

- A polynomial is separable if it has no repeated roots in a splitting field.
- An algebraic element is separable if its minimal polynomial is; an algebraic extension is separable if all its elements are.

Perfect field. F is perfect if every irreducible polynomial over F is separable (equivalently, every algebraic extension of F is separable).

Algebraically closed; algebraic closure.

- K is algebraically closed if every non-constant polynomial in $K[x]$ has a root in K (equivalently splits in K).
- An algebraic closure $\overline{F}$ of F is an algebraic extension of F that is algebraically closed.

Galois group, fixed field.

- $\mathrm{Gal}(E/F)$ is the group of automorphisms of E fixing F pointwise.
- For $H \leq \mathrm{Aut}(E)$, the fixed field is $E^H = \{a \in E : \sigma(a) = a \ \forall \sigma \in H\}$.

Galois extension. A finite extension E/F is Galois if $|\mathrm{Gal}(E/F)| = [E : F]$; equivalently E is the splitting field over F of a separable polynomial; equivalently E/F is normal (an irreducible polynomial over F with one root in E splits in E) and separable.

Roots of unity.

- An n-th root of unity is a root of $x^n - 1$; it is primitive if its order is exactly n.
- In $\mathbb{C}$ they form the cyclic group $\mu_n = \langle \zeta_n \rangle$, $\zeta_n = e^{2\pi i/n}$, with $\varphi(n)$ primitive ones.

2. Main results

Theorem (Splitting fields)

- Every non-constant $f \in F[x]$ of degree n has a splitting field, unique up to an isomorphism fixing F, and $[E : F] \leq n!$.
- The degree of the splitting field is not the degree of f.

Theorem (Derivative test)

- f has a repeated root iff $\gcd(f, f') \neq 1$.
- An irreducible f is inseparable iff $f' = 0$, which can happen only in characteristic p, with $f(x) = g(x^p)$.

Theorem (Perfect fields)

- Every field of characteristic 0 is perfect.
- A field of characteristic p is perfect iff the Frobenius map $a \mapsto a^p$ is onto.
- Every finite field is perfect.
- $\mathbb{F}_p(t)$ is not perfect: $x^p - t$ is irreducible (Eisenstein at t) and $x^p - t = (x - \sqrt[p]{t})^p$ in a splitting field.

Theorem (Primitive element)

- Every finite separable extension is simple, $E = F(\gamma)$.
- So every finite extension of a field of characteristic 0 or of a finite field is simple.
- A finite extension is simple iff it has finitely many intermediate fields (Steinitz).
- Non-example: $\mathbb{F}_p(s, t)/\mathbb{F}_p(s^p, t^p)$ has degree p^2 and is not simple.

Theorem (Algebraic closure)

- $\mathbb{C}$ is algebraically closed (Fundamental Theorem of Algebra), and $\overline{\mathbb{R}} = \mathbb{C}$.
- Every field has an algebraic closure, unique up to isomorphism.
- $\overline{\mathbb{Q}}$ (algebraic numbers) is countable with $[\overline{\mathbb{Q}} : \mathbb{Q}] = \infty$; $\mathbb{C}$ is not algebraic over $\mathbb{Q}$.
- No finite field is algebraically closed: $1 + \prod_{a \in F}(x - a)$ has no root in F.

Theorem (Basic facts on automorphisms)

- An F-automorphism maps each root of $f \in F[x]$ to a root of f; so if E is the splitting field of a degree-n polynomial, $\mathrm{Gal}(E/F)$ embeds in S_n.
- For finite E/F, $|\mathrm{Gal}(E/F)| \leq [E : F]$.
- Every automorphism fixes the prime subfield.

Theorem (Fundamental Theorem of Galois Theory)

Let E/F be finite Galois, $G = \mathrm{Gal}(E/F)$. Then $H \mapsto E^H$ and $K \mapsto \mathrm{Gal}(E/K)$ are mutually inverse, inclusion-reversing bijections between subgroups of G and intermediate fields $F \le K \le E$, with

$$[E : E^H] = |H|, \qquad [E^H : F] = [G : H].$$

K/F is Galois iff $\mathrm{Gal}(E/K) \trianglelefteq G$, and then $\mathrm{Gal}(K/F) \cong G/\mathrm{Gal}(E/K)$.

Theorem (Finite fields)

- $\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$ is cyclic of order n, generated by Frobenius $\sigma(a) = a^p$; more generally $\mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) \cong \mathbb{Z}_n$, generated by $a \mapsto a^q$.
- The subfield $\mathbb{F}_{p^d}$ ($d \mid n$) is the fixed field of $\langle \sigma^d \rangle$.
- If $f \in \mathbb{F}_q[x]$ is irreducible of degree n with root α, its roots are $\alpha, \alpha^q, \ldots, \alpha^{q^{n-1}}$, all distinct, and f splits in $\mathbb{F}_{q^n}$: every finite extension of a finite field is Galois.

Theorem (Cyclotomic extensions)

- $\mathbb{Q}(\zeta_n)$ is the splitting field of $x^n - 1$ over $\mathbb{Q}$, $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$, and $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong U(\mathbb{Z}_n)$ via $\sigma_a(\zeta_n) = \zeta_n^a$.
- It is abelian, and cyclic iff $n = 1, 2, 4, p^k, 2p^k$ (p odd prime).
- Over $\mathbb{F}_q$: $x^n - 1$ has $\gcd(n, q-1)$ roots in $\mathbb{F}_q$, and $\mathbb{F}_q$ contains a primitive n-th root of unity iff $n \mid q - 1$.

3. Standard tables

Polynomial over $\mathbb{Q}$	Splitting field	Degree	Galois group	Intermediate fields
$x^2 - d$, d not a square	$\mathbb{Q}(\sqrt{d})$	2	$\mathbb{Z}_2$	2
$x^3 - 1$, $x^6 - 1$	$\mathbb{Q}(\omega)$	2	$\mathbb{Z}_2$	2
$x^4 - 1$, $x^4 + 4$	$\mathbb{Q}(i)$	2	$\mathbb{Z}_2$	2
$(x^2 - 2)(x^2 - 3)$	$\mathbb{Q}(\sqrt{2}, \sqrt{3})$	4	$\mathbb{Z}_2 \times \mathbb{Z}_2$	5
$x^4 + 1$	$\mathbb{Q}(\zeta_8) = \mathbb{Q}(\sqrt{2}, i)$	4	$\mathbb{Z}_2 \times \mathbb{Z}_2$	5
$x^3 - 2$	$\mathbb{Q}(\sqrt[3]{2}, \omega)$	6	S_3	6
$x^4 - 2$	$\mathbb{Q}(\sqrt[4]{2}, i)$	8	D_4	10
$x^p - 1$, p prime	$\mathbb{Q}(\zeta_p)$	$p - 1$	$\mathbb{Z}_{p-1}$	$\tau(p-1)$
$x^n - 1$	$\mathbb{Q}(\zeta_n)$	$\varphi(n)$	$U(\mathbb{Z}_n)$	# subgroups of $U(\mathbb{Z}_n)$
$x^p - a$, p odd prime, a not a p-th power	$\mathbb{Q}(\sqrt[p]{a}, \zeta_p)$	$p(p-1)$	$\mathbb{Z}_p \rtimes \mathbb{Z}_{p-1}$	–
irreducible cubic, Δ a square	–	3	$A_3 \cong \mathbb{Z}_3$	2
irreducible cubic, Δ not a square	–	6	S_3	6

- For $x^3 + px + q$ the discriminant is $\Delta = -4p^3 - 27q^2$.
- The "intermediate fields" column counts F and E themselves.

4. Exam shortcuts and standard counter-examples

Exam shortcut.

- For a Galois extension, count intermediate fields by counting subgroups of the Galois group, and normal (Galois) intermediate fields by counting normal subgroups.
- A subfield of degree d over F corresponds to a subgroup of index d.

Exam shortcut.

- $|\mathrm{Aut}(\mathbb{F}_{p^n})| = n$; $|\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_{p^m})| = n/m$ for $m \mid n$; intermediate fields of $\mathbb{F}_{p^n}/\mathbb{F}_{p^m}$ correspond to d with $m \mid d \mid n$.
- $x^{p^n} - x$ is always separable (its derivative is -1).

Standard counter-examples.

- Not Galois: $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ (only the identity automorphism, degree 3); $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ (automorphism group of order 2, degree 4).
- Inseparable irreducible polynomial: $x^p - t$ over $\mathbb{F}_p(t)$. Non-perfect field: $\mathbb{F}_p(t)$.
- Algebraic but infinite: $\overline{\mathbb{Q}}/\mathbb{Q}$. Galois tower need not be Galois: $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt[4]{2})$, each step degree 2, but the whole is not Galois.
- Generators of $\mathbb{F}_{p^n}^\times$ generate the field, not conversely: $\mathbb{F}_9$ has 6 elements outside $\mathbb{F}_3$ but only $\varphi(8) = 4$ primitive elements.

5. Solved examples

Example 1. Find the degree over $\mathbb{Q}$ of the splitting field of (a) $x^4 - 2$ (b) $x^4 + 4$ (c) $x^4 - 5x^2 + 6$ (d) $x^5 - 1$ (e) $x^6 - 1$.

Solution

- (a) Roots $\pm\sqrt[4]{2}, \pm i\sqrt[4]{2}$; field $\mathbb{Q}(\sqrt[4]{2}, i)$, degree $4 \cdot 2 = 8$.
- (b) $x^4 + 4 = (x^2 - 2x + 2)(x^2 + 2x + 2)$, roots $\pm 1 \pm i$; field $\mathbb{Q}(i)$, degree 2.
- (c) $= (x^2 - 2)(x^2 - 3)$; field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, degree 4.
- (d) $\mathbb{Q}(\zeta_5)$, degree $\varphi(5) = 4$.
- (e) $x^6 - 1 = (x - 1)(x + 1)(x^2 + x + 1)(x^2 - x + 1)$; all roots lie in $\mathbb{Q}(\omega)$, degree 2.

Example 2. (a) Does $f = x^3 - 3x + 2$ have a repeated root? (b) Show $x^{p^n} - x$ is separable over $\mathbb{F}_p$. (c) Show $x^2 - t$ is irreducible but not separable over $\mathbb{F}_2(t)$.

Solution

- (a) $f' = 3x^2 - 3 = 3(x - 1)(x + 1)$ and $f(1) = 0$, so $\gcd(f, f') = x - 1$: repeated root; indeed $f = (x - 1)^2(x + 2)$.
- (b) Its derivative is $p^n x^{p^n - 1} - 1 = -1$, so $\gcd(f, f') = 1$.
- (c) t is prime in $\mathbb{F}_2[t]$, so $x^2 - t$ is Eisenstein at t, irreducible; its derivative is $2x = 0$, and $x^2 - t = (x - \sqrt{t})^2$ over a splitting field.

Example 3. Determine $G = \mathrm{Gal}(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})$ and all intermediate fields.

Solution

- The degree is 4 and the field is the splitting field of the separable $(x^2 - 2)(x^2 - 3)$, so $|G| = 4$.
- An automorphism sends $\sqrt{2} \mapsto \pm\sqrt{2}$, $\sqrt{3} \mapsto \pm\sqrt{3}$: $G = \{1, \sigma, \tau, \sigma\tau\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, with σ negating $\sqrt{2}$ and τ negating $\sqrt{3}$.
- Fixed fields: $\langle\sigma\rangle \to \mathbb{Q}(\sqrt{3})$, $\langle\tau\rangle \to \mathbb{Q}(\sqrt{2})$, $\langle\sigma\tau\rangle \to \mathbb{Q}(\sqrt{6})$.
- With $\mathbb{Q}$ and the whole field: 5 intermediate fields, all Galois over $\mathbb{Q}$ since G is abelian.
- Also $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$.

Example 4. For E, the splitting field of $x^3 - 2$ over $\mathbb{Q}$: find $\mathrm{Gal}(E/\mathbb{Q})$, the number of intermediate fields, and which are Galois over $\mathbb{Q}$.

Solution

- $E = \mathbb{Q}(\sqrt[3]{2}, \omega)$, $[E : \mathbb{Q}] = 6$, and G permutes the 3 roots faithfully, so $G \cong S_3$.
- S_3 has 6 subgroups: trivial, three of order 2, A_3, S_3.
- The order-2 subgroups fix the cubic fields $\mathbb{Q}(\sqrt[3]{2})$, $\mathbb{Q}(\omega\sqrt[3]{2})$, $\mathbb{Q}(\omega^2\sqrt[3]{2})$ (not Galois: these subgroups are not normal); A_3 fixes $\mathbb{Q}(\omega)$.
- The normal subgroups are $\{e\}, A_3, S_3$, so exactly 3 intermediate fields are Galois over $\mathbb{Q}$: E, $\mathbb{Q}(\omega)$, $\mathbb{Q}$.
- Consistently, $\Delta = -27 \cdot 4 = -108$ is not a square.

Example 5. Find the Galois group of (a) $x^3 - 3x + 1$ (b) $x^3 - 4x + 2$ over $\mathbb{Q}$.

Solution

- (a) Irreducible (± 1 are not roots).
- $\Delta = -4(-3)^3 - 27 = 108 - 27 = 81 = 9^2$: group $A_3 \cong \mathbb{Z}_3$; the splitting field has degree 3 (generated by any one root).
- (b) Eisenstein at 2, so irreducible.
- $\Delta = -4(-4)^3 - 27 \cdot 4 = 256 - 108 = 148$, not a square: group S_3, splitting field of degree 6.

Example 6. (a) For $\mathbb{Q}(\zeta_{12})$ find the Galois group, its quadratic subfields and the number of intermediate fields. (b) How many intermediate fields does $\mathbb{Q}(\zeta_{7})/\mathbb{Q}$ have? (c) Find $\mathbf{Gal}(\mathbb{F}_{3^{12}}/\mathbb{F}_{3^4})$ and the number of fields K with $\mathbb{F}_4 \subseteq K \subseteq \mathbb{F}_{2^{12}}$.

Solution

- (a) $G \cong U(\mathbb{Z}_{12}) = \{1,5,7,11\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, order $\varphi(12) = 4$.
- Since $\zeta_{12} = (\sqrt{3}+i)/2$, $\mathbb{Q}(\zeta_{12}) = \mathbb{Q}(\sqrt{3}, i)$, with quadratic subfields $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\omega)$; 5 intermediate fields in all.
- (b) $G \cong \mathbb{Z}_6$ has $\tau(6) = 4$ subgroups: $\mathbb{Q}$, $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\cos\tfrac{2\pi}{7})$ (degree 3), $\mathbb{Q}(\zeta_7)$.
- (c) Cyclic of order $12/4 = 3$, generated by $a \mapsto a^{81}$.
- $K = \mathbb{F}_{2^d}$ with $2 \mid d \mid 12$: $d = 2, 4, 6, 12$, so 4 fields.